Preventive Treatment of Migraine in Adults: CGRP Monoclonal Antibodies, Medication-Overuse Headache, and Non-Pharmacological Interventions

A PICO-Based Systematic Review with Three-Database Search and LLM-Assisted Screening (2016–2026)

Robust Literature Review Pipeline

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Abstract

Background. Migraine is a leading cause of disability worldwide. Traditional oral preventives (beta-blockers, topiramate, amitriptyline) are limited by modest efficacy, poor tolerability, and the risk of medication-overuse headache (MOH). The decade 2016–2026 saw the arrival of calcitonin gene-related peptide (CGRP)–targeted monoclonal antibodies and gepants.

Objective. To synthesise the 2016–2026 evidence on preventive treatment of migraine in adults across four pre-specified PICO questions — CGRP-targeted efficacy, safety, medication-overuse headache management, and non-pharmacological/behavioural interventions — with emphasis on the difficult-to-treat patient who has failed two or more oral preventives.

Methods. For each PICO, three databases (PubMed, Scopus, Embase) were searched from 2016 onward, identifying 12,338 records. After de-duplication, a per-article quality gate (CiteScore 3.0, SJR 0.5, or curated Q1/Q2 allowlist), DOI validation of all eligible records, and two-stage screening (deterministic title+abstract pre-screen followed by independent abstract-level large-language-model adjudication of the top-ranked records per PICO), 75 unique studies were included (plus 5 background references; 80 total), with 22 pivotal trials force-included.

Results. Across pivotal randomised trials, CGRP monoclonal antibodies reduced monthly migraine days by roughly 1.5–2.5 days more than placebo and approximately doubled 50% responder rates (e.g., erenumab 140 mg ~50% vs ~27% in episodic migraine). In patients with two-to-four prior preventive failures (LIBERTY, CONQUER, FOCUS), absolute responder rates were lower but the placebo-subtracted benefit and number needed to treat (~4–6) were favourable because placebo response was low. Network meta-analyses and the active-comparator HER-MES trial (50% responders 55.4% vs 31.2% for topiramate) position CGRP monoclonal antibodies as more effective and far better tolerated than oral preventives, with serious-adverse-event rates indistinguishable from placebo and markedly lower treatment discontinuation. Cardiovascular/blood-pressure and constipation signals are largely confined to post-marketing pharmacovigilance and to the receptor-targeting antibody erenumab, supporting a ligand-targeting antibody in patients with relevant comorbidity. In MOH, withdrawal alone is insufficient; CGRP monoclonal antibodies are effective without mandatory prior withdrawal and promote reversion from chronic to episodic migraine. Non-pharmacological interventions — aerobic exercise, cognitive behavioural therapy, relaxation, biofeedback, mindfulness, acupuncture, neuromodulation and sleep optimisation — provide adjunctive benefit.

Conclusions. For adults with frequent, disabling migraine and prior preventive failure or medication overuse, CGRP-targeted therapy is an evidence-based, guideline-concordant early option; antibody choice can be individualised by comorbidity, and pharmacologic prevention is best combined with acute-medication reduction and behavioural care.

1 Introduction

Migraine is among the most prevalent and disabling neurological disorders worldwide, imposing a burden that is disproportionately concentrated in the working-age population and in women. The Global Burden of Disease (GBD) Study 2023 confirms that headache disorders, of which migraine is the principal driver, remain a leading contributor to global disability across the period 1990–2023[1]. Migraine consistently ranks among the foremost causes of years lived with disability (YLDs) and, among young and middle-aged adults in particular, is one of the single largest sources of non-fatal health loss[1]. Because the disorder peaks during the decades of greatest occupational and family responsibility, its societal cost extends well beyond the clinic to lost productivity, impaired educational attainment, and substantial caregiver strain. This combination of high prevalence, a chronic relapsing course, and concentration in economically active individuals explains why migraine continues to dominate the neurological disability burden despite decades of therapeutic progress.

The clinical spectrum of migraine ranges from episodic migraine, conventionally defined by fewer than 15 headache days per month, to chronic migraine, defined by at least 15 headache days per month with migrainous features on at least 8 days. The transition from episodic to chronic disease—chronification—is associated with escalating disability, greater medication use, and a markedly higher likelihood of psychiatric comorbidity and cognitive complaints[2]. Disease severity and treatment response are routinely quantified using validated, patient-reported instruments such as the Migraine Disability Assessment (MIDAS) scale and the six-item Headache Impact Test (HIT-6), alongside disease-specific quality-of-life measures such as the Migraine-Specific Quality of Life questionnaire (MSQoL). These tools capture the functional consequences of migraine that headache-day counts alone cannot convey, and improvements in HIT-6 and MSQoL scores have become standard secondary endpoints in contemporary preventive trials[3]. The magnitude of day-to-day impairment—patients in pivotal chronic-migraine cohorts report baseline monthly migraine-day counts approaching 19–20 days[4]—underscores why effective prevention, rather than acute treatment alone, is central to reducing the disorder's overall impact. Even in episodic migraine, trial populations typically carry a baseline of roughly 8–9 migraine days per month[5], a frequency sufficient to compromise occupational performance and erode quality of life. The predominance of women among those affected, together with the disorder's onset in early adulthood, means that migraine-related disability accrues across precisely the years in which work, education, and family demands are greatest.

For much of the modern era, migraine prophylaxis has relied on oral agents repurposed from other indications, principally beta-blockers (e.g., propranolol), the antiepileptic topiramate, and the tricyclic antidepressant amitriptyline. Network meta-analytic evidence supports the efficacy of these agents over placebo, but only with moderate certainty, and at a meaningful cost in tolerability: valproate and amitriptyline are associated with high-certainty evidence of substantial adverse events leading to discontinuation, while topiramate and beta-blockers carry moderate-certainty evidence of treatment-limiting adverse events[6]. The clinical consequence of this poor tolerability is starkly illustrated by head-to-head data: in a randomized active-controlled trial, only 10.6% of patients discontinued erenumab because of adverse events compared with 38.9% of those receiving topiramate (odds ratio 0.19; 95% CI 0.13–0.27)[7]. Such attrition translates into low real-world adherence, frequent preventive failure, and a substantial population of patients who cycle through multiple oral agents without durable benefit. Compounding this problem is the risk of medication-overuse headache (MOH), the most common secondary headache disorder, which arises from and perpetuates the frequent use of acute medications; in pivotal chronic-migraine cohorts more than half of patients (587 of 1130; 51.9%) met criteria for baseline medication overuse[3]. MOH simultaneously worsens disability and renders conventional prevention less effective, creating a self-reinforcing cycle that is difficult to break with withdrawal alone[8].

The decade spanning 2016–2026 has witnessed a fundamental shift in the preventive treatment of migraine. Monoclonal antibodies (mAbs) targeting calcitonin gene-related peptide (CGRP) or its receptor—erenumab, fremanezumab, galcanezumab, and eptinezumab—together with the orally administered small-molecule CGRP-receptor antagonists (gepants) emerged from a robust phase 3 trial programme and rapidly entered routine practice. Across episodic and chronic migraine, these agents produced consistent, placebo-adjusted reductions in monthly migraine days and meaningful gains in 50% responder rates[4,5,9,10]. Crucially, network meta-analyses with formal GRADE appraisal now place CGRP-pathway mAbs and gepants at the head of the therapeutic hierarchy, providing high-certainty evidence of efficacy combined with adverse-event rates not significantly different from placebo—a safety-efficacy profile judged superior to all traditional oral preventives[6,11]. Their tolerability advantage is most decisive in patients who have already failed conventional therapy: dedicated trials in difficult-to-treat populations with documented failure of two to four prior preventive classes demonstrated clear benefit for galcanezumab (between-group difference -3.1 monthly migraine days vs placebo; 95% CI -3.9 to -2.3)[12], fremanezumab (placebo-adjusted reductions of -3.1 to -3.5 days)[13], and erenumab, in which 30% of treated patients achieved a 50% reduction in monthly migraine days versus 14% on placebo (odds ratio 2.7; 95% CI 1.4–5.2)[14]. CGRP-targeted therapy is also effective against medication overuse itself, both directly and as a component of combination strategies that outperform abrupt withdrawal alone[3,8]. This evidence has prompted guideline repositioning, exemplified by the American College of Physicians' clinical guideline and its companion network meta-analysis on pharmacologic prevention of episodic migraine[15,16] and by the European Headache Federation's guidance on the use of CGRP-pathway monoclonal antibodies[17]. In parallel, evidence for non-pharmacological and behavioral approaches—cognitive behavioral therapy, relaxation training, and mindfulness-based therapies—has matured, with systematic review showing these may each reduce attack frequency, albeit on a low strength-of-evidence base[18].

Despite this progress, important questions remain unresolved. The comparative efficacy and long-term safety of individual CGRP-targeting agents, their role in patients with concurrent medication overuse, and the place of behavioral and non-pharmacological interventions within an integrated preventive strategy are not yet fully defined—particularly for the difficult-to-treat patient who has failed two or more oral preventives. This systematic review therefore synthesizes the 2016–2026 evidence with the following objectives:

1. To evaluate the efficacy of CGRP-targeted monoclonal antibodies for the prevention of episodic and chronic migraine, including responder rates and reductions in monthly migraine days relative to placebo and to active comparators.
2. To appraise the safety and tolerability of CGRP-pathway mAbs and gepants, encompassing treatment-emergent and serious adverse events, discontinuation rates, and cardiovascular and vascular safety signals.
3. To assess the management of medication-overuse headache, comparing withdrawal strategies, oral prophylaxis, and CGRP-targeted and combination therapies for reducing monthly headache days and promoting reversion from overuse.
4. To examine the evidence for non-pharmacological and behavioral interventions—including cognitive behavioral therapy, relaxation, mindfulness, and exercise—as preventive options or adjuncts.
5. To synthesize evidence specific to the difficult-to-treat patient with at least two prior oral-preventive failures, with the aim of informing treatment sequencing and clinical decision-making in this high-unmet-need population.

2 Methods

2.1 Protocol, framework, and reporting

This review was conducted and reported in accordance with PRISMA 2020 and was structured around four pre-specified **PICO** questions. No prospective protocol was registered. Each PICO was searched, screened, and synthesised independently; results were then integrated into a single narrative.

PICO	Population	Intervention	Comparator	Outcome
P1 Efficacy	Adults with episodic or chronic migraine (incl. 2 prior preventive failures)	CGRP monoclonal antibodies (erenumab, galcanezumab, fremanezumab, eptinezumab); gepants	Placebo or active oral preventive	Monthly migraine-day reduction; 50% responder rate
P2 Safety	Adults with migraine receiving CGRP-targeted therapy	CGRP monoclonal antibodies / gepants	Placebo or oral preventive	Serious adverse events; cardiovascular/blood pressure; constipation; discontinuation
P3 MOH	Adults with chronic migraine and medication-overuse headache	CGRP mAbs and/or withdrawal/preventive strategies	Usual care / withdrawal alone / placebo	Headache-day reduction; reversion to episodic / no overuse

PICO	Population	Intervention	Comparator	Outcome
P4 Non-pharmacological	Adults with migraine	Exercise, CBT, biofeedback, relaxation, mindfulness, acupuncture, neuromodulation, sleep/lifestyle	Control / usual care / waitlist	Headache/migraine-day reduction; disability

2.2 Data sources and search strategy

For each PICO, three databases were searched (publications from 1 January 2016 onward): **PubMed** (NCBI E-utilities), **Scopus** (Elsevier), and **Embase**. PubMed searches used PICO-specific Boolean queries restricted to a curated allowlist of 28 high-impact (Q1/Q2) headache, neurology, pain, and general-medicine journals; Scopus searches used the corresponding `TITLE-ABS-KEY(...)` queries without journal restriction (uncapped). The Embase channel was implemented as a MEDLINE-subject (`subj=MEDI`) Scopus query and is therefore a high-overlap proxy rather than a fully Embase-indexed search (true Embase indexing requires a separate subscription); it was capped at 400 records per PICO. Citation counts and journal CiteScore/SJR metrics were obtained from the Scopus serial-title API (1,069 unique ISSNs resolved and cached). Open-access status and DOI resolution were verified through Unpaywall and the doi.org handle API.

2.3 Eligibility, quality gate, and DOI validation

Records were merged across databases and de-duplicated by DOI and title. A **per-article quality gate** then retained records with **CiteScore 3.0 OR SJR 0.5 OR** membership of the curated Q1/Q2 allowlist (pre-specified pivotal trials were exempt). Every record passing the quality gate and the relevance pre-screen (i.e., all eligible records) underwent **DOI validation** via the doi.org handle API and Unpaywall; records with an unresolvable DOI were excluded as an anti-hallucination safeguard.

2.4 Two-stage screening (deterministic + LLM adjudication)

Screening proceeded in two stages. **Stage 1** applied a deterministic title-plus-abstract filter that removed off-topic/non-headache, paediatric-only, migraine-variant-specific (vestibular, hemiplegic, aura-provocation), non-migraine-entity, acute-treatment-only, comorbidity/genetic-association-only, and letter/editorial records. Because the eligible pools remained large, records were then ranked (study design, recency, topical centrality, citation metrics) and the **top 150 per PICO** were advanced to **Stage 2**, in which an independent large-language-model judge (Claude Haiku) adjudicated each record at the abstract level, returning an include/exclude decision, a 0–3 relevance score, the study design, and an extracted quantitative key finding. Twenty-two pre-specified pivotal trials/landmark analyses were force-included across the relevant PICOs.

2.5 Study selection and synthesis

Within each PICO, LLM-included records with relevance 2 (plus force-included pivotal trials) were ranked and capped to form the final evidence set (P1 = 26, P2 = 18, P3 = 20, P4 = 18). After cross-PICO de-duplication this yielded **75 unique studies**, supplemented by 5 background/context references for the introduction, pathophysiology, and diagnosis sections (**80 references total**). Structured quantitative data (MMD reductions, responder rates, NNT/NNH, p-values, confidence intervals, dosing) were synthesised narratively per PICO. Because trials differ in population, baseline frequency and placebo response, between-trial absolute comparisons are interpreted cautiously and not pooled.

2.6 PRISMA flow (per PICO)

Stage	P1 Efficacy	P2 Safety	P3 MOH	P4 Non-pharm
Records identified (PubMed / Scopus / Embase)	3,234 (735/2,099/400)	2,372 (461/1,511/400)	1,420 (280/740/400)	5,312 (1,479/3,433/400)
After de-duplication	2,136	1,551	768	4,340
After per-article quality gate	1,175	852	438	2,097

Stage	P1 Efficacy	P2 Safety	P3 MOH	P4 Non-pharm
Eligible after title+abstract pre-screen	981	682	399	1,537
DOI-validated eligible	981	682	399	1,537
Shortlisted for LLM adjudication	150	150	150	150
LLM-included (relevance 2)	80	110	110	68
Final included (prioritised)	**26**	**18**	**20**	**18**

Total records identified across the four PICOs: 12,338. All eligible records had resolvable DOIs (no DOI-validation exclusions). Final unique included studies: 75 (+5 background references).

2.7 Limitations of the method

Inclusion was prioritised rather than exhaustive: after relevance pre-screening, only the top-ranked 150 records per PICO were advanced to LLM adjudication, so lower-ranked eligible records were not individually adjudicated. LLM screening was single-pass (no dual independent human reviewer) and no formal pooled meta-analysis or quantitative risk-of-bias scoring was undertaken. The Embase channel was a MEDLINE-subject Scopus proxy, and Scopus/Embase share an underlying corpus, so the three databases are not fully independent. The search was limited to 2016 onward and to records retrievable through these APIs, and some recent (2025–2026) systematic reviews were assessed at the abstract level.

3 Pathophysiology and the Rationale for CGRP-Targeted Prevention

3.1 The trigeminovascular system and the genesis of migraine pain

Migraine is now understood as a disorder of the brain rather than a primary disease of the cranial vasculature, with the trigeminovascular system serving as the final common pathway for headache pain. Nociceptive afferents arising from the first (ophthalmic) division of the trigeminal nerve innervate the pain-sensitive intracranial structures, most notably the dura mater and its large vessels, the pial and cerebral arteries, and the venous sinuses. The cell bodies of these pseudounipolar neurons reside in the trigeminal ganglion, and their central projections synapse within the trigeminocervical complex, the functional continuum formed by the caudal trigeminal nucleus (trigeminal nucleus caudalis) and the dorsal horns of the upper cervical segments. From this second-order relay, nociceptive traffic ascends through the brainstem and thalamus to the somatosensory cortex, while modulatory influences descend from the hypothalamus, periaqueductal gray, and rostral pons. Activation of perivascular trigeminal afferents triggers the orthodromic transmission of pain signals and, in parallel, the antidromic release of vasoactive neuropeptides at peripheral nerve terminals, producing neurogenic inflammation characterized by vasodilation, plasma protein extravasation, and mast-cell degranulation. This integrated peripheral and central circuit explains the throbbing, lateralized quality of migraine pain and its sensitivity to manipulation of perivascular signaling molecules.

Cortical spreading depression (CSD) is widely regarded as the electrophysiological substrate of the migraine aura. CSD is a slowly propagating wave of near-complete neuronal and glial depolarization that advances across the cortical mantle at approximately 2 to 5 mm per minute, followed by a prolonged wave of suppressed electrical activity. The transient depolarization produces the positive sensory phenomena of aura, such as scintillating scotomata, while the ensuing suppression corresponds to the negative symptoms that follow. Beyond generating aura, CSD is thought to provide a mechanistic link to headache itself: the massive transmembrane ionic shifts and the release of protons, potassium, glutamate, nitric oxide, and arachidonic acid metabolites into the interstitial space can activate and sensitize the perivascular trigeminal afferents that overlie the cortex, thereby coupling the cortical event to downstream trigeminovascular nociception.

3.2 CGRP biology and the related PACAP target

Among the neuropeptides released by trigeminal afferents, calcitonin gene-related peptide (CGRP) has emerged as the molecule of central pathophysiologic and therapeutic importance. CGRP is a 37-amino-acid peptide generated by alternative splicing of the calcitonin gene and is densely expressed throughout the trigeminovascular system. It is one of the most potent endogenous vasodilators known and simultaneously functions as a pronociceptive neuromodulator, facilitating synaptic transmission within the trigeminocervical complex and promoting peripheral and central sensitization. Critically, CGRP concentrations rise in cranial venous outflow during spontaneous and experimentally induced migraine attacks and normalize as the attack resolves or after successful treatment, and exogenous CGRP infusion can provoke migraine-like headache in susceptible individuals. These observations established CGRP as both a biomarker of trigeminal activation and a causal participant in attack generation, providing the empirical foundation on which contemporary preventive strategies have been built; indeed, the central role of CGRP in migraine pathophysiology is explicitly invoked as the rationale for these therapies[10]. The pituitary adenylate cyclase-activating polypeptide (PACAP) pathway represents a closely related, partially independent signaling axis: like CGRP, PACAP is a vasoactive neuropeptide that can trigger migraine-like attacks when infused, and it is now being pursued as an additional, complementary target for prevention.

3.3 Central sensitization and chronification

Repetitive activation of the trigeminovascular system can drive a transition from episodic to chronic migraine through mechanisms of central sensitization. Sustained nociceptive input enhances the excitability and lowers the activation threshold of second-order neurons in the trigeminocervical complex and of third-order thalamic neurons, manifesting clinically as cutaneous allodynia and the expansion of receptive fields. With ongoing sensitization, attacks become more frequent and more refractory, and the disorder may progress to a chronic, treatment-resistant state. Glial and neuroinflammatory processes are increasingly implicated in this maintenance phase, whereby activated microglia and the release of proinflammatory mediators perpetuate neuronal hyperexcitability within central pain pathways. These chronification mechanisms are clinically salient because chronic migraine, frequently complicated by medication overuse, is the population in which the disease burden is greatest and in which CGRP-targeted prevention has demonstrated meaningful benefit, including reversion from medication overuse[3,8].

3.4 Mechanistic rationale for anti-CGRP therapies

The convergence of evidence implicating CGRP in attack generation provided a rational, mechanism-driven target for prophylaxis, and two pharmacologic classes now exploit it. The first comprises monoclonal antibodies directed at the CGRP axis, which differ in a fundamental and clinically relevant way according to where they act. Erenumab is a fully human monoclonal antibody that binds the canonical CGRP receptor, thereby acting as a receptor-targeting (receptor-blocking) agent that prevents endogenous CGRP from engaging its receptor[7,10,19]. By contrast, galcanezumab, fremanezumab, and eptinezumab are ligand-targeting antibodies that bind the CGRP peptide itself and sequester it before it can reach the receptor[5,9,12,13]. Galcanezumab and fremanezumab are humanized antibodies that selectively bind CGRP, and both reduce monthly migraine days even in patients in whom multiple prior preventive classes had failed. The distinction between receptor and ligand blockade is mechanistically important because the CGRP receptor is also activated by related peptides, so the two antibody strategies may differ in the breadth of signaling they interrupt and in their downstream physiological consequences. The second class consists of the gepants, orally administered small-molecule antagonists of the CGRP receptor; together with the antibodies, these agents constitute the targeted CGRP-pathway armamentarium that network meta-analyses have grouped as CGRP(receptor) monoclonal antibodies and gepants and have identified as having a favorable balance of efficacy and tolerability among preventive options[6,11].

3.5 Vascular-safety rationale arising from CGRP physiology

Because CGRP is among the most powerful endogenous vasodilators and is released within the cardiovascular and cerebrovascular beds, its physiology raises a theoretical mechanistic concern: chronic pharmacologic interruption of CGRP signaling could, in principle, impair a compensatory vasodilatory reserve that protects tissues during ischemic or hypertensive stress. This consideration applies with particular force to receptor-blocking agents, since selective blockade of the canonical CGRP receptor removes the principal route through which the peptide exerts its vasodilatory effect[19]. The same vasodilatory biology that makes CGRP an attractive antinociceptive target therefore also frames the cardiovascular and cerebrovascular safety questions that accompany sustained CGRP blockade. These mechanistic considerations motivate the dedicated appraisal of vascular and blood-pressure outcomes presented in the safety section that follows, where the clinical evidence bearing on this theoretical risk is examined in detail.

4 Diagnosis, Disability Assessment, and Medication-Overuse Headache

4.1 Diagnostic Framework

Accurate diagnosis is the foundation of rational preventive treatment, and the contemporary standard is the third edition of the International Classification of Headache Disorders (ICHD-3). Under ICHD-3, *migraine without aura* requires at least five attacks lasting 4 to 72 hours (untreated or unsuccessfully treated), each accompanied by at least two of four characteristic pain features—unilateral location, pulsating quality, moderate-to-severe intensity, and aggravation by routine physical activity—together with at least one of nausea/vomiting or the combination of photophobia and phonophobia. *Migraine with aura* is defined by recurrent, fully reversible focal neurological symptoms (most often visual, but also sensory, speech/language, motor, brainstem, or retinal) that develop gradually over 5 or more minutes, last 5 to 60 minutes, and are typically followed by headache. These syndromic criteria distinguish migraine from tension-type headache and from secondary headaches that require separate evaluation.

A clinically decisive distinction is drawn between *episodic migraine* and *chronic migraine*, because monthly attack frequency drives both prognosis and treatment selection. Chronic migraine is operationally defined as headache occurring on 15 or more days per month for more than 3 months, of which at least 8 days per month meet criteria for migraine (or respond to migraine-specific acute treatment). Episodic migraine encompasses everything below this threshold and is frequently subdivided into low-frequency (fewer than 8 monthly migraine days) and high-frequency (8 to 14 monthly migraine days, with fewer than 15 total headache days) phenotypes. These cut points are not arbitrary: pivotal trials enrolled and stratified participants on exactly these grounds, with chronic-migraine studies recruiting patients averaging roughly 18 to 20 monthly headache days[4] and dedicated trials in treatment-resistant populations explicitly classifying patients as low-frequency episodic, high-frequency episodic, or chronic by these day-count rules[12].

4.2 Disability and Impact Instruments

Because headache frequency alone underestimates the lived consequences of migraine, validated patient-reported instruments are used to quantify disability and guide the decision to initiate prophylaxis. The Migraine Disability Assessment (MIDAS) questionnaire captures the number of days over the preceding 3 months on which work, household, and social activity were lost or substantially impaired; scores of 21 or more denote severe disability (grade IV), 11 to 20 moderate disability, 6 to 10 mild, and 0 to 5 little or none. The six-item Headache Impact Test (HIT-6) measures the functional and psychosocial impact of headache on a scale of 36 to 78, with scores of 60 or above signifying severe impact and 50 or above indicating a substantial effect on daily life. Both instruments are embedded as efficacy endpoints in modern preventive trials, where treatment-associated reductions in MIDAS and HIT-6 corroborate reductions in migraine-day counts[3,20].

These measures complement, rather than replace, frequency-based thresholds for initiating preventive therapy. Contemporary guidance generally supports offering pharmacologic prophylaxis to patients with frequent attacks—commonly operationalized as approximately 4 or more migraine days per month, or fewer attacks when disability is high or acute treatment is inadequate, contraindicated, or overused. The 2025 American College of Physicians clinical guideline endorses initiating preventive pharmacotherapy in the outpatient setting for adults with episodic migraine to reduce attack frequency and the attendant disability burden[15], and the European Headache Federation similarly frames monthly migraine-day frequency and disability as the principal triggers for prophylaxis, including with monoclonal antibodies targeting the calcitonin gene-related peptide (CGRP) pathway[17]. Reflecting this convention, the registration trials enrolled patients with at least 4 migraine days per month[7].

4.3 Medication-Overuse Headache

Medication-overuse headache (MOH) is a secondary headache disorder defined by ICHD-3 as headache occurring on 15 or more days per month in a patient with a pre-existing primary headache, developing as a consequence of regular overuse of one or more acute or symptomatic headache medications for more than 3 months. The diagnosis is drug-class specific, and the day-count thresholds differ accordingly. These thresholds are not merely definitional; they are the same cut points used to identify medication overuse as a baseline characteristic in chronic-migraine prevention trials[3].

Acute medication class	Overuse threshold	Minimum duration
Simple analgesics (acetaminophen, aspirin, other NSAIDs)	15 days/month	>3 months
Triptans	10 days/month	>3 months
Ergotamine	10 days/month	>3 months
Combination analgesics	10 days/month	>3 months
Opioids	10 days/month	>3 months
Multiple drug classes combined	10 days/month	>3 months

The pathophysiology of medication overuse is incompletely understood but is thought to involve maladaptive central sensitization and dysregulation of descending pain-modulatory pathways. Chronic exposure to acute medications appears to lower the threshold for cortical spreading depression, up-regulate pronociceptive signaling (including within the trigeminovascular and CGRP systems), and engage reward-related and behavioral conditioning circuits, producing a state of facilitated pain transmission that perpetuates frequent headache.

4.4 Epidemiology and the Bidirectional Burden

MOH is the most common secondary headache disorder and a major contributor to the global burden of headache, which remains among the leading causes of years lived with disability worldwide[1]. It is closely intertwined with chronic migraine: medication overuse is present in roughly half of patients enrolled in chronic-migraine cohorts—51.9% in one large pivotal trial population[3]—and is a principal modifiable driver of the transition from episodic to chronic migraine. The relationship is bidirectional. Increasing attack frequency prompts escalating use of acute medication, while that escalating use itself promotes chronification and sustains the high-frequency state, creating a self-reinforcing cycle that is difficult to interrupt with acute treatment alone. Randomized and meta-analytic evidence confirms that this population is both prevalent and therapeutically challenging, and that withdrawal of the overused agent without concurrent prevention is frequently insufficient to break the cycle[8,21,22].

4.5 Why Medication Overuse Reframes Preventive Strategy

The recognition of MOH fundamentally reframes preventive planning. First, it changes the diagnostic target: a patient presenting with 15 or more monthly headache days may have chronic migraine, MOH, or—most often—both, and effective management must address acute-medication exposure rather than simply layering on additional symptomatic drugs. Second, it changes outcome expectations, because overuse can blunt the apparent efficacy of prophylaxis and confound the interpretation of trial results unless overuse status is measured and analyzed. Reassuringly, modern preventive agents retain efficacy in the presence of overuse and can themselves reduce acute-medication consumption: CGRP-pathway monoclonal antibodies produced clinically meaningful reductions in headache days and shifted substantial proportions of overusing patients back to non-overuse status[3,23,24]. Network meta-analytic data further indicate that combination strategies pairing withdrawal or restriction of the overused medication with active prevention—including anti-CGRP therapies and nerve blocks—outperform withdrawal alone[8,22]. These observations establish the rationale, developed in the subsequent section, for an integrated MOH-management approach in which initiating or optimizing prevention is central to reversing chronification rather than an afterthought to acute-medication withdrawal.

5 Efficacy of CGRP-Targeted Therapies for Migraine Prevention

Therapies directed against calcitonin gene-related peptide (CGRP) or its receptor—four injectable monoclonal antibodies (mAbs; erenumab, galcanezumab, fremanezumab, eptinezumab) and the oral small-molecule gepants (atogepant, rimegepant)—constitute the first migraine-specific preventive class. This section synthesizes the pivotal placebo-controlled randomized clinical trials (RCTs), the dedicated trials in difficult-to-treat populations, and the indirect comparative evidence from network meta-analyses (NMAs), with reference to monthly migraine-day (MMD) reduction and the 50% responder rate. Throughout, it should be emphasized that absolute outcomes are not directly comparable across trials: baseline migraine frequency, placebo response, endpoint definitions (migraine days vs headache days), and population enrichment differ substantially, so cross-trial numerical contrasts are descriptive rather than statistically valid. Only head-to-head RCTs and formal NMAs support comparative inference.

5.1 Pivotal RCTs in Episodic Migraine

The foundational episodic-migraine (EM) program established a consistent, dose-dependent benefit over placebo. In the 6-month phase 3 STRIVE trial of erenumab (n = 955; baseline 8.3 MMD), MMDs fell by 3.2 days with 70 mg and 3.7 days with 140 mg versus 1.8 days with placebo ($P < 0.001$ for each), and 50% responder rates reached 43.3% and 50.0% versus 26.6%[25]. The smaller 12-week ARISE trial (n = 570) tested erenumab 70 mg, yielding a more modest 2.9-day reduction versus 1.8 days for placebo (treatment difference −1.0 day; 95% CI −1.6 to −0.5) and a 50% responder rate of 39.7% versus 29.5% (OR 1.59; 95% CI 1.12–2.27; $P = 0.010$)[10]. The galcanezumab EVOLVE program was larger and more durable: EVOLVE-1 (n = 858 intent-to-treat) demonstrated superiority of both 120 mg and 240 mg monthly over placebo across 6 months[20], and EVOLVE-2 (n = 915) reduced MMDs by 4.3 days (120 mg) and 4.2 days (240 mg) versus 2.3 days for placebo (differences −2.0 and −1.9 days), with both doses superior for all key secondary endpoints including 50%, 75%, and 100% responder rates[9]. The fremanezumab HALO-EM trial (n = 875) showed that monthly dosing (225 mg) and a single higher quarterly dose (675 mg) both outperformed placebo, reducing migraine days by 4.0 and 3.1 days respectively over 12 weeks; notably, patients with prior failure of two or more preventive classes were excluded, restricting generalizability to a more treatment-responsive population[5]. Across these EM trials, absolute MMD reductions clustered around 3–4 days and 50% responder rates around 40–50%, with placebo responses of 1.8–2.3 days—differences that, while modest in absolute terms, were highly consistent.

5.2 Pivotal RCTs in Chronic Migraine

Chronic-migraine (CM) populations carry far higher baseline burden (typically 15 headache and 8 migraine days monthly), and the pivotal CM trials reported outcomes as monthly headache or migraine days reduced from these elevated baselines. The phase 2 erenumab CM trial (n = 667; randomized 3:2:2 to placebo [n = 286], 70 mg [n = 191], or 140 mg [n = 190]; baseline ~18 MMD) was the first to show a significant MMD reduction over placebo: both erenumab doses reduced MMDs by 6.6 versus 4.2 for placebo (difference −2.5; 95% CI −3.5 to −1.4; $p < 0.0001$), with 50% responder rates of 40% (70 mg) and 41% (140 mg) versus 23% for placebo (OR 2.2 and 2.3, respectively), supporting advancement to phase 3 dosing[26]. The galcanezumab REGAIN trial (n = 1113; baseline 19.4 monthly headache days) reduced monthly headache days by 4.8 (120 mg) and 4.6 (240 mg) versus 2.7 for placebo, providing Class I evidence of superiority[4]. The fremanezumab

HALO-CM trial (n = 1130; baseline ~13 headache days) reduced headache days by 4.6 (monthly) and 4.3 (quarterly) versus 2.5 for placebo ($P < 0.001$)[27]. The larger absolute reductions in CM than EM reflect the higher baselines rather than greater relative efficacy, underscoring why absolute cross-population comparison is misleading.

5.3 Difficult-to-Treat Migraine: 2 Prior Preventive Failures

The most clinically decisive evidence comes from dedicated phase 3b RCTs enrolling patients with documented failure of two to four (or up to four) preventive classes—the population in whom oral preventives have already proven inadequate and for whom guideline-endorsed CGRP-mAb use is concentrated. These trials are critical because enriched-failure populations typically show lower placebo responses, sharpening the estimated treatment effect.

The erenumab LIBERTY trial (n = 246; EM with 2–4 prior failures) randomized patients to 140 mg or placebo and met its primary endpoint: 30.3% of erenumab-treated patients achieved a 50% MMD reduction at weeks 9–12 versus 13.7% on placebo (OR 2.7; 95% CI 1.4–5.2; $P = 0.002$)[14]. Its 3-year open-label extension demonstrated durability and accrual of benefit: 240/246 (97.6%) entered the extension and, among those with valid data, 52.3% (79/151) were 50% responders at week 168, with mean MMD reduction of approximately 4.4 days; importantly, 40.8% of placebo-treated non-responders converted to responders after switching to erenumab, indicating that early non-response does not preclude later benefit[28]. The galcanezumab CONQUER trial (n = 462 randomised across 64 sites; EM [58%] or CM [42%] with 2–4 category failures) randomized patients 1:1 to 120 mg monthly (240 mg loading) or placebo (n = 232 vs 230) for 3 months: galcanezumab reduced monthly migraine headache days by 4.1 versus 1.0 for placebo (between-group difference −3.1; 95% CI −3.9 to −2.3; $p < 0.0001$), with a 50% responder rate of 37.7% versus 13.3% (OR 3.9; 95% CI 2.7–5.7) and a 75% rate of 14.5% versus 3.3% (both $p < 0.0001$)[12]. The fremanezumab FOCUS trial—the largest dedicated failure-population RCT (n = 838; EM [39%] or CM [61%]; documented failure of two to four classes)—randomized patients 1:1:1 to quarterly fremanezumab (n = 276), monthly fremanezumab (n = 283), or placebo (n = 279): monthly migraine days fell by 4.1 (monthly) and 3.7 (quarterly) versus 0.6 for placebo (differences −3.5 [95% CI −4.2 to −2.8] and −3.1 [−3.8 to −2.4]; both $p < 0.0001$), with 50% responder rates of 34% for both fremanezumab regimens versus 9% for placebo (OR 5.8 for each; $p < 0.0001$)[13]. Collectively, LIBERTY, CONQUER, and FOCUS provide the strongest direct support for CGRP mAbs in the 2-failure population, with consistent benefit despite the more refractory case mix.

This evidence is corroborated and ranked by NMA. In a Bayesian NMA restricted to prior-failure patients (7 studies, 3052 patients), galcanezumab 240 mg ranked first for both MMD reduction (mean difference [MD] −4.40 days; 95% credible interval [CrI] −7.60 to −1.19) and 50% response (RR 4.18; 95% CrI 2.63–6.67), and CGRP-ligand mAbs were superior to the receptor mAb erenumab for achieving 50% response[29]. The most recent and comprehensive failure-population NMA (18 RCTs, 7281 participants; 13 contributing to the network and GRADE-rated) found that, versus placebo, fremanezumab (MD −3.30; 95% CI −4.11 to −2.49), eptinezumab (−3.35; −4.38 to −2.32), galcanezumab (−2.73; −3.43 to −2.03), atogepant (−2.30; −3.47 to −1.13), and erenumab (−2.20; −2.72 to −1.68) were among the most effective for reducing MMDs (low certainty), and all increased the likelihood of 50% response—fremanezumab (RR 3.98; 95% CI 2.40–6.59), atogepant (2.80; 1.73–4.54), erenumab (2.56; 2.01–3.26), eptinezumab (2.35; 1.61–3.42), and galcanezumab (1.94; 1.52–2.48); the 50%-response estimates were of moderate certainty for erenumab, eptinezumab, and galcanezumab and low certainty for fremanezumab and atogepant[30].

5.4 Comparative Effectiveness and Network Meta-Analyses

In the absence of definitive head-to-head efficacy RCTs among the mAbs, NMAs provide the principal comparative evidence, and their conclusions are broadly concordant on class-level superiority over placebo while diverging on internal ranking. A large NMA of 74 trials (32,990 patients) found high-certainty evidence that CGRP(r) mAbs, gepants, and topiramate increase the 50% responder proportion over placebo, with moderate-certainty benefit for beta-blockers, valproate, and amitriptyline—but the latter two carried high-certainty harms causing discontinuation, reinforcing the favorable benefit–tolerability balance of the CGRP class[6]. A phase 3-only NMA (19 studies, 14,584 participants spanning EM and CM) concluded that all CGRP agents (atogepant, rimegepant, erenumab, eptinezumab, fremanezumab, galcanezumab) decreased mean MMDs and increased 50%, 75%, and 100% responder rates versus placebo, with the partial exception of eptinezumab 30 mg[31]. In CM specifically, an NMA of 13 RCTs (5634 participants) ranked single-dose eptinezumab 300 mg best for MMD reduction (MD −2.60 days; 95% CI −4.43 to −0.77) and fremanezumab best for 50% response (OR 2.96; 95% CI 2.20–3.97), with erenumab 140 mg most effective for reducing acute medication days[32]. These rankings should be read cautiously: they rest on indirect comparisons across heterogeneous populations and dosing, the rank-best agent differs by outcome and analysis, and no formal head-to-head RCT confirms any single mAb as superior.

Direct comparative data against active oral preventives strengthen the case for the class on tolerability rather than separating efficacy. In the active-controlled phase 4 HER-MES trial ($n = 777$), erenumab (70/140 mg) versus optimally dosed topiramate (50–100 mg/day) showed markedly lower discontinuation for adverse events (10.6% vs 38.9%; OR 0.19; 95% CI 0.13–0.27) alongside a favorable efficacy profile[7]. The pragmatic 12-month APPRAISE trial randomized EM patients with 1–2 prior failures 2:1 to erenumab versus non-specific oral preventives, with the primary endpoint of completing the assigned treatment while achieving 50% MMD reduction at month 12 favoring erenumab—an endpoint capturing the combined value of sustained efficacy and adherence[33].

5.5 Gepants and Real-World Effectiveness

Oral gepants extend CGRP-receptor antagonism to a more accessible delivery route. The phase 3b ELEVATE trial established efficacy of atogepant 60 mg once daily versus placebo for EM in patients failed by two to four conventional oral classes—the first dedicated gepant trial in a refractory population ($n = 315$ randomised; 309 in the primary off-treatment hypothetical estimand population)—reducing MMDs by 4.2 versus 1.9 for placebo across the 12-week period (least-squares mean difference -2.4; 95% CI -3.2 to -1.5; $p < 0.0001$), with a 50% responder rate of 51% versus 18% (OR 4.8; 95% CI 2.9–8.1; $p < 0.0001$)[34]. Rimegepant 75 mg every other day was evaluated for prevention in a phase 3 Japanese RCT ($n = 484$ efficacy; baseline ~9 MMD), meeting its primary endpoint with a statistically significant MMD reduction versus placebo (difference -1.1 days), confirming a preventive signal for an oral as-needed/preventive agent outside the United States[35]. Gepants are increasingly positioned as effective comparators within the failure-population NMAs cited above, where atogepant achieved point estimates competitive with the mAbs for both MMD reduction and 50% response[30].

Beyond regulatory trials, effectiveness has been examined in comorbid and pragmatic settings. In the UNITE trial ($n = 353$) of patients with migraine and comorbid major depressive disorder, monthly fremanezumab 225 mg reduced MMDs significantly versus placebo over 12 weeks, demonstrating retained efficacy in a psychiatrically complex group often excluded from pivotal trials[36]. Together, these data support efficacy that extends across episodic and chronic migraine, treatment-refractory patients, and clinically representative comorbid populations.

5.6 Summary of Pivotal Trials

Trial (citekey)	Drug, dose	Population (n)	MMD/MHD reduction vs placebo	50% responders
STRIVE[25]	Erenumab 70/140 mg	EM (955)	−3.2 / −3.7 vs −1.8 d	43.3% / 50.0% vs 26.6%
ARISE[10]	Erenumab 70 mg	EM (570)	−2.9 vs −1.8 d (Δ −1.0)	39.7% vs 29.5%
EVOLVE-1[20]	Galcanezumab 120/240 mg	EM (858)	Superior to placebo	Superior to placebo
EVOLVE-2[9]	Galcanezumab 120/240 mg	EM (915)	−4.3 / −4.2 vs −2.3 d	Superior (all key secondary)
HALO-EM[5]	Fremanezumab 225 mg monthly / 675 mg quarterly	EM (875)	−4.0 / −3.1 d	Superior to placebo
Erenumab phase 2[26]	Erenumab 70/140 mg	CM (667)	−6.6 / −6.6 vs −4.2 MMD (Δ −2.5)	40% / 41% vs 23%
REGAIN[4]	Galcanezumab 120/240 mg	CM (1113)	−4.8 / −4.6 vs −2.7 MHD	Superior to placebo
HALO-CM[27]	Fremanezumab 675 mg q / 225 mg m	CM (1130)	−4.3 / −4.6 vs −2.5 MHD	Superior to placebo
LIBERTY[14]	Erenumab 140 mg	EM, 2–4 failures (246)	—	30.3% vs 13.7% (OR 2.7)
LIBERTY 3-yr[28]	Erenumab 140 mg (OLE)	EM, 2–4 failures (240)	~−4.4 d (wk 168)	52.3% at wk 168
CONQUER[12]	Galcanezumab 120 mg	EM/CM, 2–4 failures (462)	−4.1 vs −1.0 MMD (Δ −3.1)	37.7% vs 13.3% (OR 3.9)

Trial (citekey)	Drug, dose	Population (n)	MMD/MHD reduction vs placebo	50% responders
FOCUS[13]	Fremanezumab 225 mg m / 675 mg q	EM/CM, 4 failures (838)	−4.1 / −3.7 vs −0.6 (Δ −3.5 / −3.1)	34% / 34% vs 9% (OR 5.8)
ELEVATE[34]	Atogepant 60 mg QD	EM, 2–4 failures (315)	−4.2 vs −1.9 (Δ −2.4)	51% vs 18% (OR 4.8)

Across the totality of evidence, CGRP-targeted therapies deliver reproducible, clinically meaningful reductions in migraine frequency and consistently higher responder rates than placebo in episodic and chronic migraine, with particularly important benefit demonstrated in the difficult-to-treat population of patients in whom two or more prior preventives have failed. NMAs corroborate class-level superiority and a favorable tolerability profile relative to older oral agents, but their internal rankings are sensitive to outcome choice, population, and indirectness, and no head-to-head RCT has established a definitive efficacy hierarchy among the individual antibodies or gepants. Any apparent absolute advantage of one agent over another drawn from separate trials should therefore be interpreted with caution.

6 Safety and Tolerability of CGRP-Targeted Therapies

A central rationale for the rapid adoption of calcitonin gene-related peptide (CGRP)-targeted preventives is their favorable tolerability relative to conventional oral agents. This section synthesizes the safety evidence for P2—adults with migraine receiving CGRP monoclonal antibodies (mAbs) or gepants compared with placebo or oral preventives—across serious adverse events (SAEs), cardiovascular and blood-pressure outcomes, specific signals (notably constipation and pharmacovigilance reports), and treatment discontinuation. Throughout, randomized controlled trial (RCT) evidence is distinguished from spontaneous-reporting pharmacovigilance data, the latter being hypothesis-generating rather than confirmatory.

6.1 Overall Safety and Serious Adverse Events Versus Placebo

Across the pooled RCT literature, CGRP-targeted therapies are not associated with an excess of serious adverse events relative to placebo. An early meta-analysis of 10 RCTs (5817 participants) evaluating four mAbs in episodic migraine found no significant association between mAb treatment and SAEs (relative risk [RR] 1.13, 95% CI 0.74–1.72, $p = 0.56$), overall withdrawals (RR 0.90, 95% CI 0.77–1.05, $p = 0.17$), withdrawals due to adverse events (RR 1.39, 95% CI 0.95–2.04, $p = 0.09$), or any adverse event (RR 0.99, 95% CI 0.90–1.10, $p = 0.90$); the dominant treatment-emergent signals were injection-site pain (RR 1.32, 95% CI 1.02–1.71, $p = 0.03$) and injection-site erythema (RR 1.55, 95% CI 1.17–2.05, $p < 0.01$)[37]. This null SAE finding has been reproduced consistently. A network meta-analysis of 19 phase 3 RCTs (14 584 patients) comparing mAbs and gepants reported no significant differences in SAEs between any active treatment and placebo, with eptinezumab associated with the lowest odds of treatment-emergent adverse events (TEAEs) and SAEs[11]. Differences emerge chiefly in the milder TEAE category: atogepant 120 mg (odds ratio [OR] 2.22, 95% CI 1.26–3.91) and galcanezumab 240 mg (OR 1.63, 95% CI 1.33–2.00) carried the largest odds of TEAEs versus placebo[11]. For atogepant specifically, a meta-analysis of 4 RCTs (2813 participants) confirmed a modest TEAE excess (relative risk [RR] 1.10, 95% CI 1.02–1.21) without dose-dependent worsening[38]. Dose-ranging analyses of individual agents corroborate the benign SAE profile, with no evident difference from placebo for eptinezumab across 30-, 100-, and 300-mg regimens[39] and consistent safety across erenumab doses: in a meta-analysis of 8 RCTs (4860 patients), any-AE rates did not differ from placebo at 7 mg (RR 0.9, 95% CI 0.7–1.2), 21 mg (RR 1.0, 95% CI 0.8–1.2), 28 mg (RR 0.9, 95% CI 0.7–1.1), 70 mg (RR 1.0, 95% CI 0.9–1.0), or 140 mg (RR 1.0, 95% CI 0.9–1.1), and SAEs (1.5% vs 1.4% for placebo) and AE-related discontinuations (1.1% vs 1.0%) were likewise comparable across all doses[40]. Broader meta-analytic syntheses reach the same conclusion: a meta-analysis of 13 RCTs (6979 patients) pooling erenumab 70 mg, fremanezumab 225 mg, and galcanezumab 120 mg found no significant difference versus placebo in any adverse event (odds ratio [OR] 1.10, 95% CI 0.99–1.21), treatment-related adverse events (OR 1.12, 95% CI 0.72–1.72), or SAEs (OR 0.96, 95% CI 0.63–1.46)[41]. In populations with prior preventive failure or medication-overuse headache—groups of particular clinical relevance—safety remained comparable to placebo: a network meta-analysis of treatment-refractory patients found no differences in TEAE or SAE incidence across comparisons[29], and an updated meta-analysis of moderate-to-high-dose eptinezumab and erenumab in chronic migraine with medication-overuse headache found no substantial difference in TEAEs leading to discontinuation and similar SAE rates versus placebo[24]. Long-term data extend this reassurance: a meta-analysis of studies with at least 12 months of mAb exposure estimated a pooled discontinuation rate of 23% for any reason but only approximately 3% attributable to adverse events, although all contributing studies carried a severe risk of bias by design[42].

6.2 Cardiovascular and Blood-Pressure Considerations

Because CGRP is a potent vasodilator and a putative protective mediator during ischemia, sustained pathway blockade raised a theoretical concern for adverse cardiovascular and blood-pressure effects, with the question of whether receptor-targeting (erenumab) differs from ligand-targeting agents remaining biologically plausible but unresolved. The most rigorous RCT-derived evidence comes from an integrated analysis of vascular safety across four double-blind placebo-controlled erenumab studies and their open-label extensions (2443 patients in the controlled phase). Adverse-event incidence was similar across placebo and erenumab groups regardless of acute medication use or baseline vascular risk factors; hypertension adverse events were reported in 0.9% (placebo), 0.8% (erenumab 70 mg), and 0.2% (erenumab 140 mg) of patients[19]. Of 18 adjudicated cases, 4 events—2 deaths and 2 vascular events—were positively classified as cardiovascular in origin, and all 4 occurred during open-label erenumab treatment without a placebo comparator, underscoring the difficulty of causal attribution[19]. Following the post-approval addition of hypertension to erenumab prescribing information, a dedicated systematic review and meta-analysis examined blood-pressure outcomes and found no significant association with elevated systemic blood pressure (systolic mean difference 0.86 mmHg, 95% CI -1.02 to 2.73; $p = 0.370$)[43]. The authors nonetheless emphasized considerable fragility in the available evidence and advised individualized risk-benefit assessment in patients with multiple hypertension risk factors[43]. On balance, randomized evidence does not demonstrate a clinically meaningful blood-pressure or cardiovascular hazard, but the populations studied largely excluded patients with established cardiovascular disease, limiting generalizability to higher-risk groups.

6.3 Specific Signals: Constipation, Infection, and Pharmacovigilance Reports

Beyond aggregate measures, several agent-specific signals warrant attention. Constipation is the most clinically salient gastrointestinal effect and is most pronounced with CGRP-receptor blockade. For atogepant, meta-analysis demonstrated a more than two-fold increase in constipation versus placebo

(RR 2.55)[38]. For the receptor-targeting mAb erenumab, RCT-level data confirm a constipation signal: in the dose meta-analysis, constipation was reported in 3.9% of erenumab-treated patients versus 1.6% of placebo recipients[40], and in chronic migraine a systematic review found that both erenumab (RR 2.92, 95% CI 1.44–5.89) and atogepant (RR 3.32, 95% CI 1.61–6.88) probably increase the likelihood of constipation versus placebo, whereas the ligand-targeting fremanezumab showed little or no difference[44]. Infection risk has also been scrutinized given CGRP's role in immune signaling: a PRISMA-Harms meta-analysis of 37 placebo-controlled RCTs (22 518 patients) found a small overall increase in infection risk (RR 1.08, 95% CI 1.01–1.14; number needed to harm [NNH] 287), driven in individual analyses only by galcanezumab at clinical doses (RR 1.13, 95% CI 1.02–1.25; NNH 77) and eptinezumab at higher doses (RR 1.23, 95% CI 1.04–1.45; NNH 24)[45]. These effect sizes are small and the high NNH for the pooled estimate suggests limited clinical impact for most patients[45]. Pharmacovigilance data add a complementary but lower-tier perspective. A systematic review of 30 spontaneous-reporting studies drawing on the FDA Adverse Event Reporting System, WHO VigiBase, and EudraVigilance identified consistent signals of disproportionate reporting including injection-site reactions, alopecia, and constipation, along with reports raising the question of vasoconstrictive phenomena such as reversible cerebral vasoconstriction syndrome and Raynaud phenomenon[46]. These disproportionality signals must be interpreted cautiously: spontaneous reporting cannot establish incidence or causation, is subject to reporting bias, and is hypothesis-generating only. Reassuringly, the rarer vasoconstrictive signals have not materialized as excess events in controlled trials[11,19], and these signals should not be overstated relative to the consistent randomized evidence.

6.4 Tolerability Versus Oral Preventives and Clinical Implications

The tolerability advantage of CGRP-targeted agents is clearest when contrasted with oral preventives, whose discontinuation rates are driven by adverse events. Because head-to-head trials are scarce, indirect comparisons have been informative: an indirect meta-analysis of episodic migraine RCTs comparing mAbs with topiramate calculated number-needed-to-harm and likelihood-to-help-or-harm metrics favoring the mAbs on adverse-event-related discontinuation[47], and a network meta-analysis of chronic migraine that incorporated topiramate and onabotulinumtoxinA found favorable acceptability (dropout) for CGRP mAbs[32]. The active-comparator paradigm exemplified by the HER-MES design—a direct erenumab-versus-topiramate trial—has been pivotal in demonstrating substantially lower discontinuation due to adverse events with the mAb, a contrast that anchors the tolerability narrative for these agents. A comprehensive systematic review of 43 RCTs (14 725 participants) in chronic migraine reinforced this, finding with moderate-certainty evidence that galcanezumab probably reduces all-cause dropout versus placebo (RR 0.52, 95% CI 0.33–0.83) while the CGRP-targeted class showed little or no difference in discontinuation due to adverse events versus placebo; by contrast, botulinum toxin probably increased AE-related discontinuation (RR 3.36, 95% CI 1.75–6.45)[44]. Tolerability also supports persistence: in the LIBERTY 3-year open-label extension, 70% of participants with multiple prior preventive failures completed long-term erenumab treatment, consistent with durable acceptability[28]. The network meta-analysis literature in difficult-to-treat populations similarly identifies CGRP agents among the most effective and acceptable options[30]. For clinical practice, these data support several implications. CGRP-targeted therapy is a tolerable option for most patients, including those who have failed multiple oral preventives. For patients with pre-existing constipation, ligand-targeting mAbs or careful monitoring may be preferable to erenumab or atogepant given the receptor-pathway constipation signal[38,44]. For patients with cardiovascular risk factors, randomized evidence does not contraindicate use, but the exclusion of high-risk patients from trials, the residual fragility of blood-pressure data[43], and the open-label adjudicated cardiovascular events[19] justify individualized risk-benefit discussion and blood-pressure surveillance, particularly with erenumab. Overall, the safety profile of CGRP-targeted preventives is favorable and well characterized for SAEs, with the principal uncertainties confined to rare pharmacovigilance signals and to populations underrepresented in randomized trials.

7 Medication-Overuse Headache Management and Comparison With Oral Preventives

Medication-overuse headache (MOH) is the most common secondary headache disorder and a frequent companion of chronic migraine (CM), affecting roughly 40% to 56% of CM populations enrolled in pivotal preventive trials—for example, 40.2% of the PROMISE-2 cohort and 51.9% of the HALO CM cohort carried a baseline MOH or medication-overuse diagnosis[3,48]. This PICO addresses adults with CM plus MOH and asks whether calcitonin gene-related peptide (CGRP) monoclonal antibodies (mAbs) and/or withdrawal and preventive strategies, compared with usual care, withdrawal alone, or placebo, reduce headache days and promote reversion from chronic to episodic migraine and to no medication overuse.

7.1 Managing MOH: Withdrawal Alone Versus Combination Strategies

Historically, treatment hinged on withdrawal of the overused acute medication, but the necessity of concurrent prophylaxis was long contested[22]. The most informative synthesis is the network meta-analysis (NMA) by Koonalintip and colleagues, which pooled 16 randomized controlled trials (RCTs) enrolling 3,000 participants and ranked strategies by reduction in monthly headache days[8]. Combination approaches dominated: abrupt withdrawal plus oral prevention plus greater occipital nerve block achieved the largest reduction (−10.6 days/month; 95% CI, −15.03 to −6.16), and restriction of the overused medication plus oral prevention plus CGRP therapy ranked second (−8.47 days/month; 95% CI, −12.78 to −4.15). Single-modality strategies—oral prevention, anti-CGRP(receptor) therapy, or botulinum toxin alone—were less effective than these combinations, supporting concurrent prophylaxis over withdrawal alone[8].

A complementary NMA by Kong and colleagues screened 8,248 publications and analyzed 28 RCTs against the outcomes of responder rate, reversion to no medication overuse (nMO), and reductions in headache and acute-medication frequency[22]. Topiramate emerged as broadly beneficial—responder-rate odds ratio (OR) 4.93, headache frequency weighted mean difference (WMD) −5.53, and acute-medication frequency WMD −6.95, with fewer adverse events than placebo (OR 0.20). Fremanezumab, galcanezumab, and botulinum toxin type A (BTA) improved responder rates (ORs 3.46–3.07, 2.95, and 2.57, respectively), while eptinezumab, fremanezumab, and BTA were superior to placebo for reversion to nMO (ORs up to 2.75, 1.87, and 1.55)[22]. Non-pharmacologic adjuncts also have a role: the MIND-CM RCT showed that mindfulness added to usual care benefited patients with CM and MOH[49]. Pediatric evidence remains thin—a systematic review of 16 studies (only one RCT) in children and adolescents found no guideline-grade data[50].

7.2 CGRP Monoclonal Antibodies in MOH Without Mandatory Withdrawal

A central modern question is whether CGRP-targeted therapy can succeed without enforced detoxification. The accumulated evidence increasingly answers yes. A 2025 systematic review and meta-analysis of three RCTs (769 patients) found that standard-to-high-dose eptinezumab and erenumab roughly doubled the odds of a 50% reduction in monthly migraine days (MMDs) versus placebo (OR 2.43; 95% CI, 1.68–3.51; $P < .00001$), without a meaningful excess of treatment-emergent adverse events leading to discontinuation[24]. A dedicated erenumab (70 mg) meta-analysis (five RCTs, 875 patients; mean age 42.7 years) reported significant reductions in acute headache-medication days (four studies; mean difference [MD] −1.72; 95% CI, −2.81 to −0.62; $P = .002$) and MMDs (three studies; MD −1.88; 95% CI, −2.68 to −1.07; $P < .001$), and a higher likelihood of a 50% reduction in MMDs at 3 months (four studies; RR 1.49; 95% CI, 1.26–1.77; $P < .001$), with overall adverse events comparable to placebo (RR 1.02; 95% CI, 0.92–1.14) but constipation the main tolerability signal (RR 1.43; 95% CI, 1.17–1.76)[23]. Giri and colleagues, pooling topiramate, BTA, and CGRP mAb RCTs in CM-MOH, found the mAbs significantly superior to placebo across all outcomes, with a 50% responder OR of 2.90 (95% CI, 2.23–3.78), whereas BTA reduced headache frequency by only 1.92 days/month (95% CI, −2.68 to −1.16)[21].

Reversion outcomes are clinically pivotal. In the HALO CM subgroup, monthly and quarterly fremanezumab reverted 53% to 55% of patients with baseline overuse to no overuse (quarterly 111/201 [55.2%]) and roughly tripled the 50% responder rate versus placebo (34.8%–39.4% vs 13.8%)[3]. A broad CGRP systematic review reported that anti-CGRP mAbs halve MMDs in 27.6% to 61.4% of CM patients, drive chronic-to-episodic conversion in about 40.9%, and eliminate medication overuse in 29% to 88%, with obesity the principal negative predictor and no clear superiority among agents[51]. The DRAGON post hoc analysis quantified reversion directly: erenumab 70 mg increased the proportion of Asian CM patients reverting to episodic migraine, a benefit preserved across subgroups including baseline medication-overuse status[52]. Real-world durability is supported by the PREVAIL extension, in which 43 of 49 MOH patients (87.8%) achieved 50% reduction in MIDAS-derived headache days on eptinezumab over up to 2 years[53], and by PROMISE-2, where eptinezumab nearly doubled the proportion reverting to no overuse versus placebo while improving HIT-6 and other patient-reported outcomes[48]. The phase 4 erenumab trial in nonopioid MOH and the RESOLUTION trial of eptinezumab with structured patient education further establish efficacy without mandatory withdrawal[54,55]. The oral gepant atogepant extends this to a small molecule: in the phase 3 PROGRESS trial, 66.2% of participants met overuse criteria, and atogepant 30 mg twice daily reduced MMDs by a least-squares mean difference of −2.7 (95% CI, −4.0 to −1.4) versus placebo, with comparable benefit in the 60 mg once-daily arm[56].

The detox-versus-non-detox debate has thus shifted. Whereas glucocorticoids such as prednisone were historically used to ease withdrawal symptoms, a narrative review concluded that RCT evidence for steroids is inconsistent and that their role is diminished in the CGRP era[57]. Imaging data suggest medication withdrawal itself may help normalize aberrant central pain processing, providing a mechanistic rationale for combination care when withdrawal is feasible[58].

7.3 Effect of Discontinuing Therapy and Comparator Strategies

Sustained benefit appears contingent on continued treatment; the long-term PREVAIL and atogepant real-world cohorts demonstrate maintained response only while therapy continues, and patients followed after their last infusion illustrate the relevance of treatment continuity[53,59]. In a difficult-to-treat real-world atogepant cohort (100 resistant CM patients, median six prior preventive failures), 60 mg daily produced clinically meaningful MMD reductions and 50% responder rates sustained to 24 weeks[59].

7.4 Head-to-Head and Oral Preventive Comparisons

Direct comparison with oral preventives is anchored by HER-MES, a randomized, double-blind, active-controlled phase 4 trial of erenumab versus topiramate, in which erenumab was better tolerated, with markedly fewer adverse-event discontinuations[7]. Comparative NMAs and guidelines refine positioning. Kong's NMA actually ranked topiramate highly for efficacy outcomes in MOH but with a less favorable tolerability balance than expected in practice[22], while the broader comparative-effectiveness NMA of preventives reinforces the strong, consistent efficacy of CGRP-pathway drugs[6,44]. Real-world oral-preventive data temper expectations: in the prospective observational MigriNor cohort (254 patients, 51.2% chronic migraine, 18.1% with baseline overuse), candesartan and amitriptyline produced similar, modest reductions in moderate-to-severe headache days over 12 weeks (each −2.4 days/4 weeks at weeks 9–12; 95% CI, −3.0 to −1.9 and −3.1 to −1.6, respectively), with 30% responder rates of 59.6% and 57.1% and 50% responder rates of only 44.9% and 34.9%, while 22.3% to 46.8% of participants discontinued before 12 weeks and adverse drug reactions affected 77.1% (candesartan, highest acceptability) to 95.7% (topiramate, lowest acceptability)—a real-world tolerability burden that contrasts with the more favorable adverse-event-driven discontinuation profile of CGRP-pathway therapy[60]. The CandeSpartan study likewise reported 50% responder rates of only 34.9% to 36% in a population that was 55.8% MOH[61]. Fremanezumab additionally reduced acute-medication use in episodic migraine, underscoring its capacity to interrupt the overuse cycle upstream[62].

7.5 Other Pharmacologic Options and Positioning

OnabotulinumtoxinA is a validated option: a systematic review of four RCTs (1,259 patients) found BTA significantly reduced headache frequency versus placebo (MD 1.89; 95% CI, 1.11–2.67), and combining BTA with anti-CGRP therapy likely benefits resistant cases[51,63]. Repurposed agents—topiramate, candesartan, amitriptyline—retain a role where access to CGRP therapy is constrained[22,60].

Strategy	Representative evidence	Key outcome
Combination (withdrawal + oral + nerve block)	Koonalintip NMA[8]	−10.6 headache days/month
CGRP mAbs (eptinezumab/erenumab)	Nguyen meta-analysis[24]	50% MMD reduction, OR 2.43
Fremanezumab, reversion to nMO	HALO CM[3]	55.2% reverted to no overuse
Atogepant (oral gepant)	PROGRESS[56]	MMD LSMD −2.7 vs placebo
Topiramate	Kong NMA[22]	responder OR 4.93
OnabotulinumtoxinA	Lang review[63]	headache frequency MD 1.89
Oral preventives (candesartan/amitriptyline)	MigriNor[60]	modest, similar reductions

For the difficult-to-treat CM-MOH patient, the evidence favors initiating a CGRP mAb or atogepant without mandating prior detoxification, layering withdrawal, occipital nerve block, behavioral therapy, or onabotulinumtoxinA where overuse persists, and reserving oral preventives as accessible alternatives—an approach that maximizes both headache-day reduction and reversion to episodic, non-overusing disease[8,22,51].

8 Non-Pharmacological and Behavioral Interventions

Non-pharmacological and behavioral strategies are recommended adjuncts—and, for many patients, preferred first-line options—in the preventive management of migraine. They are attractive when pharmacotherapy is contraindicated, poorly tolerated, declined, or only partially effective, and they directly target the modifiable behavioral, psychological, and lifestyle factors that perpetuate attacks. This section synthesizes the evidence addressing PICO P4 (adults with migraine; non-pharmacological or behavioral interventions; control, usual care, or waitlist comparators; headache/migraine-day reduction and disability), spanning behavioral therapies, mindfulness, aerobic exercise, acupuncture, non-invasive neuromodulation, and lifestyle adjuncts. Across modalities the recurring theme is a favorable safety profile coupled with generally low-to-moderate certainty evidence, constrained by the difficulty of blinding behavioral interventions and reliance on subjective, self-reported endpoints.

8.1 Behavioral Therapies: CBT, Relaxation, and Biofeedback

The most comprehensive recent synthesis of behavioral prevention is the systematic review and meta-analysis by Treadwell and colleagues, which included 50 randomized trials in adults (77 publications, N = 6024) of predominantly multicomponent interventions combining cognitive behavioral therapy (CBT), biofeedback, relaxation training, mindfulness-based therapies, and patient education[18]. The review concluded that any of three individual components—CBT, relaxation training, or mindfulness-based therapies—may reduce migraine frequency relative to control. Importantly, most included trials were judged to be at high risk of bias, driven chiefly by potential measurement bias (unblinded, self-reported outcomes) and incomplete outcome data, so the strength of evidence was correspondingly modest. These techniques are reasonably grouped because they share a common mechanism—enhancing self-regulation of physiological arousal and the maladaptive cognitions and stress responses that trigger attacks—and because they are frequently delivered in combination.

A separate GRADE-based appraisal by Beier and colleagues reached a more tempered conclusion: across 9 RCTs of psychological treatment and 2 RCTs of manual joint mobilization, neither intervention changed the critical outcomes of headache frequency or quality of life, with overall certainty rated very low for psychological treatment (downgraded for risk of bias, inconsistency, and imprecision) and low for manual therapy (downgraded for imprecision); both nonetheless received weak recommendations as supplements to standard care, given the absence of serious adverse events and patient preferences[64]. Taken together, the literature supports CBT, relaxation, and biofeedback as safe, guideline-endorsed adjuncts with plausible benefit, but the magnitude of effect on migraine-day reduction remains uncertain and the evidence base is fragile.

8.2 Mindfulness in Chronic Migraine with Medication-Overuse Headache

Mindfulness-based interventions have attracted particular interest for the difficult population with chronic migraine (CM) complicated by medication-overuse headache (MOH). The phase-III, single-blind MIND-CM randomized controlled trial provides the strongest single-trial signal: 177 patients with CM and MOH were randomized 1:1 to treatment as usual (TaU)—comprising withdrawal from overused drugs, education, and tailored prophylaxis—versus TaU plus a six-session mindfulness program with brief daily self-practice[49]. For the primary endpoint of at least a 50% reduction in headache frequency at 12 months, the TaU + mindfulness group significantly outperformed TaU alone (78.4% vs. 48.3%), with secondary benefits reported across medication intake, quality of life, disability, and disease cost. As an add-on to standard care in a population where behavioral engagement is clinically valuable, mindfulness represents a low-risk, scalable intervention; the single-blind design and self-reported headache diary nonetheless temper the certainty of the responder estimate.

8.3 Aerobic and Physical Exercise

Aerobic exercise is among the most studied non-pharmacological options. The meta-analysis by Lemmens and colleagues pooled six studies and found a significant but small reduction in migraine frequency, with a mean decrease of approximately 0.6 ($\pm$ 0.3) migraine days per month; unpooled data suggested reductions of 20–27% in attack duration and 20–54% in pain intensity, though heterogeneity precluded pooling of these outcomes[65]. The authors graded the evidence for migraine-day reduction as moderate quality, attributing the small effect sizes partly to non-uniform outcome measures and insufficiently intense training protocols.

Comparative efficacy across exercise modalities was examined in the network meta-analysis by Woldeamanuel and Oliveira, which synthesized 21 trials (N = 1195) of moderate- and high-intensity aerobic exercise and strength/resistance training[66]. Strength training ranked highest for reducing monthly migraine frequency (mean difference approximately −3.55), followed by high- and moderate-intensity aerobic exercise (mean differences ranging roughly −2.18 to −3.13), with both exercise categories numerically outperforming the pharmacological comparators topiramate and amitriptyline in the ranking. These rankings warrant cautious interpretation: a published commentary highlighted methodological concerns, including inconsistent inclusion criteria, heterogeneous placebo/control conditions that complicate network coherence, the high or some-concern risk-of-bias ratings of the strength-training trials, and the clinical reality that exercise and preventive medication are typically combined rather than substituted[67]. A more recent network meta-analysis further explored the relative efficacy of different exercise prescriptions for migraine[68]. The overall picture is consistent: regular physical exercise modestly reduces migraine burden and is broadly recommended given its general health benefits, but precise effect estimates are unstable across analyses.

8.4 Acupuncture

Acupuncture has a substantial trial base, yet conclusions hinge critically on the comparator. The trial-sequential meta-analysis by Fan and colleagues (20 RCTs, n = 3380) found acupuncture superior to sham acupuncture for reducing migraine episodes after treatment (SMD −0.29; 95% CI −0.47 to −0.11; 11 trials, n = 1727; $I^2 = 59\%$) and for responder rate (50% reduction in monthly attacks) (RR 1.30; 95% CI 1.09 to 1.55; 9 trials, n = 1640)[69]. Against active prophylactic drugs (e.g., flunarizine, valproate, metoprolol, topiramate), the picture was more equivocal: acupuncture did not significantly outperform conventional drugs on migraine episodes after treatment (SMD −0.21; 95% CI −0.42 to 0.00; P = 0.06; 7 trials, n = 1044), but was superior on responder rate (RR 1.24; 95% CI 1.04 to 1.48; 4 trials, n = 1021) while causing fewer adverse events (RR 0.34; 95% CI 0.14 to 0.81; 6 trials)[69]. The trial-sequential analysis confirmed that the responder-rate comparisons against both sham and conventional drugs had crossed the monitoring boundary favoring acupuncture, whereas the migraine-episode comparisons had not yet reached the required information size, so the authors emphasized that more RCTs remain desirable. A more conservative recent meta-analysis by Pistoia and colleagues, comparing acupuncture against standard medical care across 15 studies, found no statistically significant difference in migraine-day frequency or pain intensity, but did observe reduced analgesic use and improved quality of life, supporting acupuncture as an adjunctive rather than stand-alone therapy[70]. Consistent with this, the GRADE-based review by Beier and colleagues (7 RCTs for acupuncture) issued a weak recommendation for acupuncture, noting possible benefit on headache frequency, intensity, quality of life, and acute-medication use, with the certainty of evidence rated low (downgraded for risk of bias and inconsistency); the review reported GRADE directions rather than pooled effect estimates[64]. Persistent heterogeneity and the lack of an adequate sham control remain the central challenges.

8.5 Non-Invasive Neuromodulation

Neuromodulation has expanded rapidly as a non-pharmacological prevention option. The systematic review and meta-analysis by Moisset and colleagues (38 articles; 31 preventive) found that, for prevention, invasive occipital nerve stimulation produced a large but heterogeneous effect, while non-invasive supraorbital transcutaneous electrical nerve stimulation, percutaneous electrical nerve stimulation, and high-frequency repetitive transcranial magnetic stimulation (rTMS) over the primary motor cortex were effective with small-to-medium effect sizes; remote electrical neuromodulation

(REN) was effective for acute treatment[71]. Focusing on chronic migraine, the meta-analysis by Zhong and colleagues (8 RCTs) reported that rTMS reduced attack frequency versus sham, with an effect size of -1.13 (95% CI -1.69 to -0.58) when the dorsolateral prefrontal cortex was targeted, although rTMS did not significantly reduce pain intensity[72]. Evidence for REN was synthesized in a dedicated systematic review and meta-analysis supporting its efficacy and favorable safety profile[73]. Non-invasive neuromodulation is therefore a promising, well-tolerated class, though device heterogeneity, small samples, and the difficulty of credible sham stimulation limit certainty.

8.6 Sleep, Lifestyle, and Other Adjuncts

Beyond discrete therapies, structured patient education and lifestyle modification—regularization of sleep, hydration, meal timing, and trigger management—are frequently bundled into multicomponent behavioral programs. In the GRADE appraisal by Beier and colleagues, patient education (5 RCTs) was associated with improved self-rated health and quality of life and an increased number of well-informed patients, supporting a weak recommendation for its use as a supplement to standard care (certainty very low, downgraded for risk of bias and imprecision), while supervised physical activity (6 RCTs) showed a possible positive impact on quality of life at longest follow-up with no change in other outcomes (certainty very low, downgraded for risk of bias, inconsistency, and imprecision)[64]. These adjuncts carry essentially no risk and reinforce self-efficacy, making them sensible components of comprehensive preventive care even where their independent effect on migraine-day counts is modest or unquantified.

8.7 Interpreting the Evidence: Blinding, Subjectivity, and Expectation

A unifying caveat applies across this literature. Behavioral, exercise, acupuncture, and neuromodulation interventions are intrinsically difficult or impossible to blind: participants and providers usually know whether active treatment is delivered, and the primary outcomes—migraine days, attack frequency, pain intensity, and disability—are self-reported. This combination invites measurement and expectation bias, the dominant reason most included trials were rated at high or unclear risk of bias[18]. Sham comparators (sham acupuncture, sham stimulation) attenuate but do not eliminate the problem, because credible blinding to a physical intervention is rarely complete, and the contrast with sham may understate real-world benefit while overstating specificity of effect[69,70]. Consequently, between-group differences are best read as the increment over an active placebo and contextual response, and apparent superiority over usual care or waitlist controls—where expectation effects are unconstrained—should be discounted accordingly. The pragmatic conclusion is that these interventions are safe, patient-valued, and probably modestly effective, justifying weak-to-moderate recommendations as adjuncts within a shared preventive strategy, while higher-quality trials with standardized outcomes and rigorous controls remain a priority.

9 Discussion

9.1 Synthesis of principal findings

Across the four PICOs, this review assembles a coherent and internally consistent picture of contemporary migraine prevention. For the first question — efficacy of calcitonin gene-related peptide (CGRP)-targeted therapy — the convergence of evidence is striking. Multiple independent network meta-analyses confirm that the monoclonal antibodies (mAbs) erenumab, fremanezumab, galcanezumab and eptinezumab, together with the gepants atogepant and rimegepant, increase the proportion of patients achieving a 50% or greater reduction in monthly migraine days relative to placebo, with the CGRP class and topiramate rated at high certainty in the most comprehensive synthesis[6,31]. Class-level efficacy is robust, but head-to-head certainty remains limited: rankings differ by outcome and by population, with eptinezumab 300 mg and erenumab 140 mg favoured for monthly migraine-day reduction and fremanezumab for responder rates in chronic migraine, while no agent is unambiguously superior overall[32]. Critically, efficacy is preserved in the population that matters most for this review — patients with prior preventive failures. The newest network meta-analysis restricted to this difficult-to-treat group reaffirms meaningful migraine-day reductions and increased responder odds for fremanezumab, eptinezumab, galcanezumab, atogepant and erenumab, albeit at low-to-moderate GRADE certainty[30], echoing earlier work in which ligand-targeting antibodies, and galcanezumab 240 mg in particular, ranked highly in prior-failure cohorts[29].

The safety PICO is reassuring while flagging genuine, if modest, signals. CGRP-pathway agents are generally well tolerated, with serious adverse events indistinguishable from placebo across syntheses[11]. This favourable profile compares well with the established oral preventives, whose tolerability constraints are well documented: amitriptyline and propranolol each raise discontinuation due to adverse events, and flunarizine likewise carries a tolerability penalty, as the European Headache Federation's critical re-appraisals of these agents make clear[74–76]. Within the CGRP class, treatment-emergent events cluster predictably by mechanism: atogepant carries an excess of constipation (RR ~2.55)[38], while higher doses of atogepant and galcanezumab show the largest odds of any treatment-emergent event[11]. Dose-finding meta-analyses of individual antibodies reinforce the broad equivalence to placebo for serious and discontinuation-related events across the licensed doses of erenumab and eptinezumab[39,40]. Two newer signals deserve attention. First, a small but statistically significant increase in infection risk across the class (RR 1.08), driven chiefly by galcanezumab and higher-dose eptinezumab, is biologically plausible given CGRP's immune role, though the number needed to harm is large[45]. Second, despite a label warning, pooled data do not demonstrate a meaningful rise in blood pressure with erenumab, though the evidence is fragile and demands individualised vigilance in patients with vascular risk[43]. Long-term (12 months) data show roughly 23% discontinuation for any reason but only ~3% for adverse events, supporting durable tolerability while underscoring the severe risk of bias inherent in observational follow-up[42]. Set against earlier comparative work showing CGRP mAbs to be at least as well tolerated as topiramate while avoiding its characteristic cognitive and paraesthetic burden[41,47,77], the safety case for the class is comparatively strong.

For medication-overuse headache (MOH), CGRP-targeted therapy is effective even without mandatory prior withdrawal. In chronic migraine complicated by MOH, eptinezumab and erenumab roughly double the odds of a 50% response versus placebo (OR 2.43) with no excess of discontinuation or serious adverse events[24]. This represents a meaningful departure from the older paradigm, in which onabotulinumtoxinA was the principal disease-specific option for chronic migraine and abrupt withdrawal of overused acute medication was often considered a prerequisite to any preventive benefit; real-world data confirm onabotulinumtoxinA's effectiveness in chronic migraine but do not resolve the management of overuse itself[78]. The current evidence instead suggests that CGRP-targeted prophylaxis can be initiated alongside, rather than after, structured acute-medication reduction. Finally, the non-pharmacological PICO supports a meaningful adjunctive role rather than a stand-alone alternative. Behavioural interventions — cognitive behavioural therapy, relaxation and mindfulness — may reduce migraine frequency and disability, though most trials are at high risk of bias[18]. Aerobic and strength exercise[66,68], acupuncture (which reduces analgesic use and improves quality of life more than it lowers migraine frequency)[69,70], and selected neuromodulation modalities including remote electrical neuromodulation and repetitive transcranial magnetic stimulation[71–73] all show small-to-moderate effects, consistent with weak recommendations for use alongside standard care[64].

9.2 Individualised, guideline-concordant decision-making

For the index patient — at least two prior oral preventive failures plus medication overuse — the evidence and contemporary guidance now align toward early CGRP-targeted therapy. The 2025 American College of Physicians guideline endorses CGRP mAbs and gepants as preventive options for episodic migraine in the outpatient setting, situating them within a value-conscious framework informed by its companion network meta-analysis[15,16]. The European Headache Federation, in its 2022 update, explicitly supports CGRP mAbs as a first-line option where reimbursement permits and after prior failures, reflecting their favourable benefit-to-harm profile[17]. In a patient who has already exhausted two oral classes and overuses acute medication, this guidance justifies moving directly to a CGRP-targeted agent rather than cycling through further oral options of comparable or lesser tolerability.

The comparative-efficacy evidence further supports this positioning. Because no oral preventive consistently outperforms the CGRP class on responder outcomes, and because the difficult-to-treat phenotype is precisely where oral agents most often disappoint, the case for early CGRP-targeted therapy rests on both efficacy retention after failure and a more favourable tolerability balance than continued oral cycling[29,30].

Antibody choice can be tailored to comorbidity. Where cardiovascular risk is a concern, the absence of a demonstrable blood-pressure effect for erenumab is reassuring but uncertain, so a ligand-targeting antibody may be preferred when vascular caution dominates[43]. Conversely, because constipation is mechanistically linked to CGRP-receptor blockade, patients with pre-existing constipation may be better served by a ligand-targeting mAb than by the receptor antagonist atogepant or by erenumab[38]. Where infection susceptibility is relevant, the class signal — modest and concentrated in galcanezumab and higher-dose eptinezumab — can inform, but should not dominate, selection[45]. Crucially, initiating a CGRP-targeted agent does not require prior detoxification: efficacy in coexisting MOH supports concurrent acute-medication reduction rather than enforced withdrawal as a prerequisite[24]. Behavioural therapy, exercise and patient education should be integrated from the outset, both to consolidate acute-medication reduction and to address the disability and psychological burden that oral monotherapy seldom resolves[18,64].

9.3 Strengths of this review

This review's methodology was designed for rigour and reproducibility. Each clinical question was framed within a formal PICO structure, ensuring that searches, screening and synthesis remained anchored to explicit populations, interventions, comparators and outcomes. The search spanned three complementary databases — PubMed, Scopus and Embase — broadening coverage beyond any single index. A per-article quality gate based on journal CiteScore restricted inclusion to higher-tier sources, and every eligible record underwent DOI validation to guarantee bibliographic integrity and traceability. Screening proceeded in two stages: a deterministic title-and-abstract pre-screen followed by independent large language model (LLM) adjudication at the abstract level, combining the consistency of rule-based filtering with contextual judgement. Pivotal randomised trials of CGRP-targeted agents were force-included to ensure that the foundational evidence base was represented regardless of ranking artefacts. Together these features yielded a transparent, defensible and current corpus spanning 2016 to 2026.

9.4 Limitations

Several limitations temper interpretation. First, inclusion was prioritised rather than exhaustive: candidate records were ranked into a shortlist before LLM adjudication, so lower-ranked but potentially eligible studies may have been missed. Second, screening relied on a single-reviewer LLM rather than dual independent human reviewers, introducing potential for systematic misclassification not captured by inter-rater statistics. Third, we performed a qualitative synthesis without a formal pooled meta-analysis or quantitative risk-of-bias scoring; we therefore rely on the GRADE ratings and bias assessments of the included syntheses, several of which judged their primary studies to be at high or severe risk of bias[18,42]. Fourth, Embase was accessed as a MEDLINE-subject Scopus proxy rather than through the native Embase interface, which may have altered retrieval. Fifth, the review was restricted to English-language publications, raising the possibility of language bias, particularly relevant for acupuncture and exercise literature concentrated in other languages. Finally, for some recent systematic reviews only abstract-level data were available, constraining the depth of extraction and the confidence of certain class- and dose-specific conclusions.

9.5 Future directions and conclusion

Priorities for future research follow directly from these gaps. Adequately powered head-to-head trials comparing CGRP mAbs with one another, and with gepants, are needed to convert class-level efficacy into agent-level guidance, especially in prior-failure and MOH populations[24,30]. Pragmatic trials should test early or first-line CGRP positioning against continued oral cycling, and rigorously evaluate combined pharmacological and behavioural pathways for the difficult-to-treat patient. Longer, lower-bias cohorts are required to characterise the infection and cardiovascular signals and to define optimal treatment duration and de-escalation[42,43,45].

In conclusion, CGRP-targeted therapies are efficacious and generally well tolerated across episodic and chronic migraine, retain benefit after prior oral failures, and remain effective in coexisting medication overuse without mandatory withdrawal. Guideline guidance now supports their early use in patients who have exhausted oral options, with antibody selection individualised by cardiovascular, gastrointestinal and infection-related comorbidity, and with behavioural and other non-pharmacological interventions integrated as adjuncts. Given the immense global burden of headache disorders[1], translating this evidence into individualised, guideline-concordant care offers substantial opportunity to reduce disability — provided remaining uncertainties around comparative efficacy and long-term safety are resolved through higher-quality, head-to-head and real-world research.

10 Appendix: PRISMA 2020 Checklist and Per-PICO Flow

10.1 Per-PICO PRISMA flow

	P1 Eff.	P2 Saf.	P3 MOH	P4 Non-ph.
Identified (PubMed/Scopus/Embase)	3,234	2,372	1,420	5,312
De-duplicated	2,136	1,551	768	4,340
Quality gate (CiteScore>=3 / SJR>=0.5 / allowlist)	1,175	852	438	2,097
Eligible (title+abstract pre-screen; all DOI-validated)	981	682	399	1,537
Shortlisted -> LLM adjudication				

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                                150       150       150      150
LLM-included (relevance >= 2)
                                 80       110       110       68
Final included (prioritised)
                                 26        18        20       18
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Total identified: 12,338. Final unique studies: 75 (+5 background references = 80). Twenty-two pivotal trials/landmark analyses were force-included.

10.2 PRISMA 2020 checklist (mapping)

#	Item	Addressed
1	Title identifies a systematic review	Title
2	Structured abstract	Abstract
3–4	Rationale & objectives (PICO)	Introduction; Methods (PICO table)
5	Eligibility criteria	Methods — Eligibility/quality gate
6	Information sources	Methods — three databases + Scopus metrics + Unpaywall/doi.org
7	Search strategy (full PICO queries)	Methods; reproduced in pipeline outputs
8	Selection process	Methods — two-stage (deterministic + LLM) screening
9–10	Data collection & items	Methods — structured extraction (MMD, responder, NNT/NNH, p, CI, dosing)
11	Risk of bias in studies	Not formally scored — quality controlled via per-article CiteScore gate; see Limitations
12	Effect measures	MMD reduction; 50%/ 75% responder; NNT/NNH; RR/OR
13	Synthesis methods	Narrative per-PICO synthesis; no pooled meta-analysis

#	Item	Addressed
14–15	Reporting bias & certainty	Discussion — narrative GRADE-style appraisal; Limitations
16	Search/selection results	Per-PICO PRISMA flow (above)
17–20	Study characteristics & results	Efficacy, Safety, MOH, Non-pharmacological sections; tables
21–22	Reporting biases & certainty	Discussion
23	Discussion (interpretation, limitations)	Discussion
24	Registration/protocol	Not registered — stated in Methods
25–26	Funding / competing interests	None (automated pipeline); none declared
27	Data/code availability	PICO queries, journal allowlist, screening rubric, and reference set provided in pipeline outputs

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